Building Block Design for Stacked Polyphase Bridge Converters for High-Voltage Traction Drives

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***Abstract*—In a stacked polyphase bridge (SPB) converter the battery voltage is shared by series-connected three-phase submodules, so the transistor voltage class is a free design variable. This paper designs the phase-leg building block of a 1200 V, 300 kW SPB traction inverter. Since the phase current is independent of the number of submodules, low-voltage GaN transistors are favored, and the turn-off overshoot at a realistic commutation-loop inductance sets 16 submodules of 75 V with 100 V devices. Four EPC2361 per switch, driven in two identical pairs by one dual-channel isolated driver each, keep the junction temperature at 69 °C and the peak drain–source voltage below 90 V. The dc link of a three-phase level is sized for the voltage ripple with the dc-bias derating of class-II MLCCs. The leg is built in a double-pulse-test version with a coaxial shunt per cell and a load-test version without, and three legs share a laminated PCB busbar. Electromagnetic extraction of the routed boards gives a commutation-loop inductance of 0.41 nH per cell without and 0.65 nH with the shunt, equal in the four parallel cells within 0.7 % and hardly coupled. The extracted gate loops are equal for the four paralleled devices of a switch but reach 20.6 nH. A double-pulse simulation with the complete extracted network confirms equal current sharing and shows that the test board needs a slower turn-on and a -1.9 V off-state to stay at 93 V.**

***Index Terms*—GaN, paralleling, parasitic inductance, stacked polyphase bridge, traction inverter.**

I. Introduction

Traction batteries of heavy-duty electric vehicles are moving to 800 V and above, which pushes two-level inverters to 1200 V transistors with large voltage steps at the machine terminals. The stacked polyphase bridge (SPB) converter [1], [2] connects N three-phase bridges in series on the dc side, each feeding one winding set of a multiphase machine. Every submodule blocks only V_{bus}/N, which allows low-voltage transistors, reduces the voltage step at the windings and the dc-link capacitance [3], [4], and allows a faulty submodule to be bypassed. Control [2], [5], dc-link stability [6], machine design [7], [8] and integration [9] have been studied, and a GaN-based SPB has been found competitive with a three-level flying-capacitor inverter [10].

The submodule itself, however, has received little attention. At 1200 V and 300 kW each leg conducts more than 200 $\mathrm{A_{rms}}$ at 75 V, so several GaN transistors must be paralleled. Paralleled low-voltage GaN stages are limited by turn-off overshoot [11], by uneven dynamic current sharing [12], [13] and by gate disturbances [14]; symmetric layouts with up to eight devices per switch have been demonstrated at 48–120 V [15], [16]. In those designs the bus voltage is given. In an SPB it is not, and the submodule voltage, the device count and the layout are coupled through the loop inductance.

This paper derives the SPB building block from the inverter specification and verifies the resulting hardware. The contributions are: 1) a selection of the device voltage class and the submodule voltage that accounts for the commutation-loop inductance; 2) a gate-drive arrangement for four paralleled GaN transistors per switch based on a current-sharing sensitivity analysis; 3) a four-cell leg board, built in a test and an operating version, and its verification by Q3D extraction of the routed copper, including the gate network, and by a double-pulse simulation with the complete extracted parasitic network.

II. Building Block Requirements

For N submodules sharing the power equally (Fig. 1a), one leg delivers $P_{\mathrm{leg}} = MV_mI_{\mathrm{ph}}\cos\varphi/(2\sqrt{2})$, so the inverter power is

$$P = 3NP_{\mathrm{leg}} = \frac{3M\,V_{\mathrm{bus}}}{2\sqrt{2}}\,I_{\mathrm{ph}}\cos\varphi. \tag{1}$$

The phase current does not depend on N: with $M = 1.15$ and $\cos\varphi = 0.9$, 300 kW at 1200 V require 228 $\mathrm{A_{rms}}$ (322 A peak) in every leg of every submodule. The number of submodules therefore only scales the voltage, and with it the number of transistors and drivers. The peak drain–source voltage is limited to 90 V for 100 V devices, and the switching frequency is set to 50 kHz to keep the submodule capacitors small.

III. Device and Submodule Voltage

Since every submodule carries the full phase current, the conduction loss of the inverter scales with $NI^2R_{\mathrm{DS(on)}}/n$ for n devices per switch, and the output-charge loss with $Nnf_{\mathrm{sw}}C_{\mathrm{oss,tr}}V_m^2$. Optimizing n gives the minimum loss

$$P_{\min} = 6\,I\,V_{\mathrm{bus}}\sqrt{f_{\mathrm{sw}}\,R_{\mathrm{DS(on)}}C_{\mathrm{oss,tr}}}, \tag{2}$$

in which the voltage class cancels to first order. Devices are thus ranked by the time constant $\tau = R_{\mathrm{DS(on)}}C_{\mathrm{oss,tr}}$ and, because conduction dominates at this current, by the on-resistance per volt of blocking capability. Among ten GaN transistors of the 100–650 V classes, the 100 V parts lead in both (Fig. 2a,b); EPC2361 (0.75 mΩ, 3 × 5 mm) has the lowest τ and 7.5 µΩ/V, three to thirty times below the higher-voltage parts. The price is a larger number of submodules (Fig. 2c),

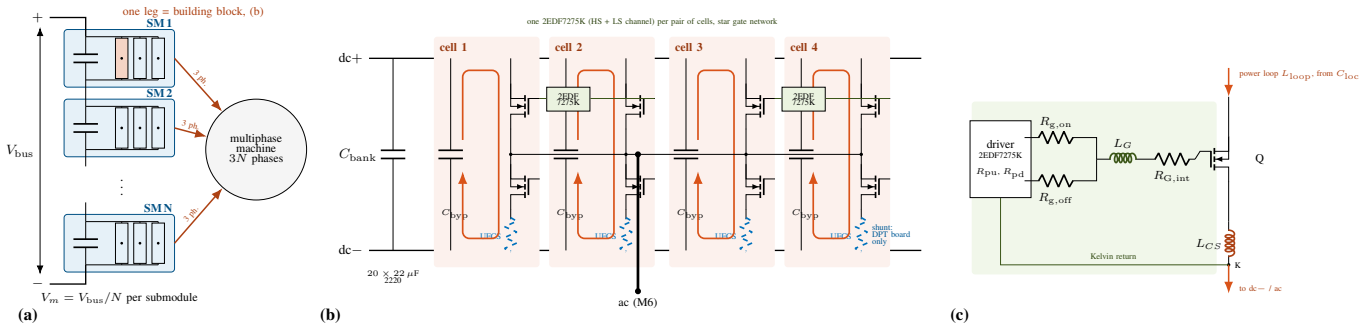


Fig. 1. (a) SPB inverter with N submodules (SM) of $V_m = V_{\rm bus}/N$. (b) Building block, one phase leg: four commutation cells (QH, QL: EPC2361) with local bypass capacitors $C_{\rm byp}$, a common capacitor bank $C_{\rm bank}$ and one ac output; one dual-channel driver serves each pair of cells. The shunt (dashed) is fitted on the test board only. (c) Gate circuit of one transistor with the common-source inductance L_{CS} up to the Kelvin pad K.

which the modular SPB accommodates. The Kelvin source pad of the EPC2361 keeps the source inductance out of the gate loop (Fig. 1c).

The submodule voltage follows from the overshoot at turn-off, $V_m + L_{\rm loop}\, di/dt \le 90\,\text{V}$. A vertical loop over a 0.1 mm dielectric reaches about 0.5 nH per cell, whereas a measurement shunt raises it to about 1 nH. With the turn-on rate of a 2EDF7275K-driven EPC2361 ($R_{g,\rm on} = 2\,\Omega$), 33 A/ns at 80 A per device, the admissible submodule voltage is 73 V at 0.5 nH and 57 V at 1 nH (Fig. 4).

An even N is needed for symmetric winding sets, so $N = 16$ with $V_m = 75\,\text{V}$ is chosen. This requires a loop close to 0.5 nH in the operating hardware; $N = 18$ would add 12.5 % transistors and conduction loss.

IV. Devices per Switch and Gate Drive

A. Number of paralleled devices

Double-pulse simulations of one EPC2361 per switch (LTspice, 75 V) bound the current per device. At 120 A the overshoot reaches 102 V with 0.5 nH, and at 80 A already 93 V with 1 nH, so the current per device is limited to about 80 A at the peak. A larger loop lowers the turn-on energy (21.8 versus 28.3 µJ at 80 A for 1 and 0.5 nH), because it takes $L_{\rm loop} di/dt$ of the voltage during the current rise, but it raises the overshoot and the turn-off energy.

The leg loss over a fundamental period is computed from these energies, with the junction temperature and $R_{\rm DS(on)}$ solved together for the thermal path of the load-test hardware: a 50 mm fan heatsink (0.65 K/W) on an aluminium spreader and, per device, 7.6 K/W from junction to heatsink, dominated by the insulating gap pad (5.3 K/W). Fig. 3 shows the result. One device per switch runs away thermally and two reach 141 °C. Three keep T_j at 84 °C but exceed 90 V. Four devices dissipate 18.0 W per leg (12.5 W conduction) at 69 °C and 40 °C ambient, with 83 V peak voltage at 0.5 nH but 93 V at 1 nH. More devices save less than 3 W per leg but add drivers. Four devices per switch are chosen, which also form two equal pairs for the gate drive below.

TABLE I
Switching-Energy Imbalance of Two Paralleled EPC2361 (LTspice, 75 V, 120 A)

Asymmetry	Imbalance	Layout measure
Power loop, L_D +0.3 nH	2 %	mirrored cells
Gate loop, L_G 2 vs. 8 nH	19 %	star gate network
No Kelvin, L_{CS} +0.2 nH	29 %	Kelvin return per device
Threshold, $V_{\rm th}$ +0.3 V	7 %	one channel per pair

B. Gate drive

Static sharing is benign because $R_{\rm DS(on)}$ rises with temperature; dynamic sharing is not. Fig. 5 and Table I show the imbalance of the switching energy for one asymmetry at a time. A 6 nH longer gate loop delays the second device by about half the current rise time and shifts most of the turn-on energy to the first. Gate-loop and common-source mismatches dominate, while a power-loop mismatch of 0.3 nH is nearly harmless. The gate paths must therefore be equal by construction. Driving all four devices of a switch from one channel is not an option: the slower gate charging raises the turn-on energy by 31 % and the peak sink current exceeds the 8 A rating. Instead, one 2EDF7275K (4/8 A, isolated high- and low-side channel, 3 ns channel mismatch) is placed between two cells that are rotated by 180° about it, and drives both gates through a star with Kelvin returns (Fig. 1b); the point symmetry makes both gate branches identical. Each isolated supply is split by a Zener diode into $+5.1\,\text{V}$ and a negative off-state voltage. The negative bias absorbs the gate bump of the off device, which the Miller current alone raises by $\Delta v_{\rm GS} \approx Q_{\rm GD}/C_{\rm GS} \approx 1.06\,\text{V}$, otherwise close to the 0.8 V minimum threshold. Its value, $-1.9\,\text{V}$, follows from the extracted gate network in Section VII and keeps the off-state ringing above the $-4\,\text{V}$ gate limit.

V. DC Link of a Three-Phase Level

The three legs of a submodule share one dc bus, so the low-frequency ripple cancels and only the switching-frequency component remains. A switching-function model with space-vector modulation gives 85 $\text{A}_{\rm rms}$ capacitor current per level and a charge excursion of 860 µC per switching period at

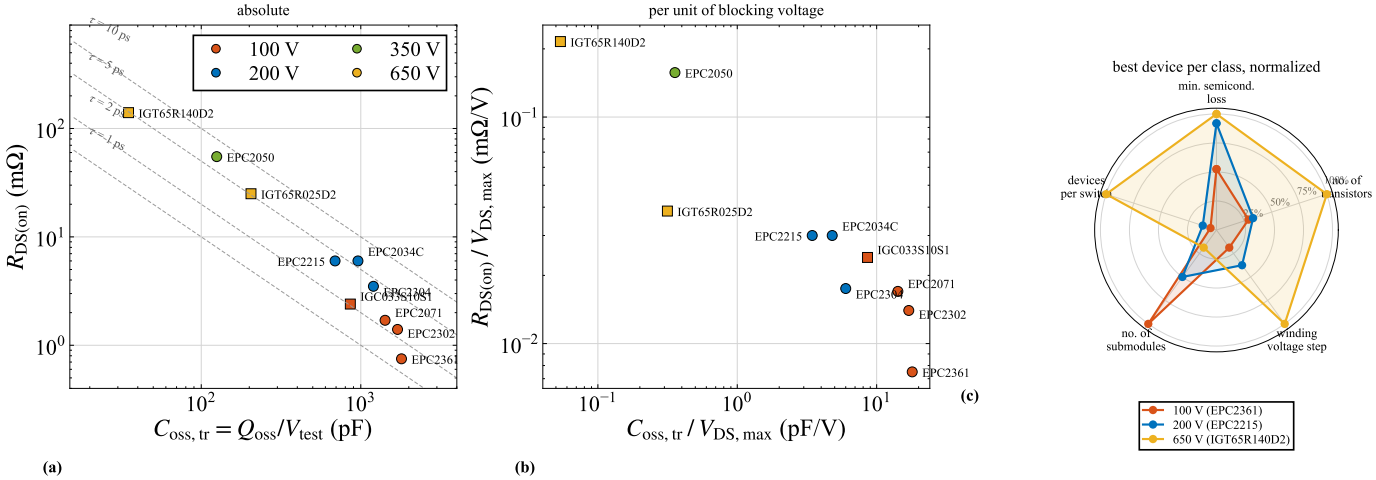


Fig. 2. Device selection for a 1200 V SPB (datasheet values). (a) $R_{\mathrm{DS(on)}}$ against $C_{\mathrm{oss,tr}}$ with lines of constant τ; (b) both per volt of blocking capability; (c) SPBs built from the best device of each class, normalized: minimum loss (2), devices per switch, submodules, voltage step and transistor count.

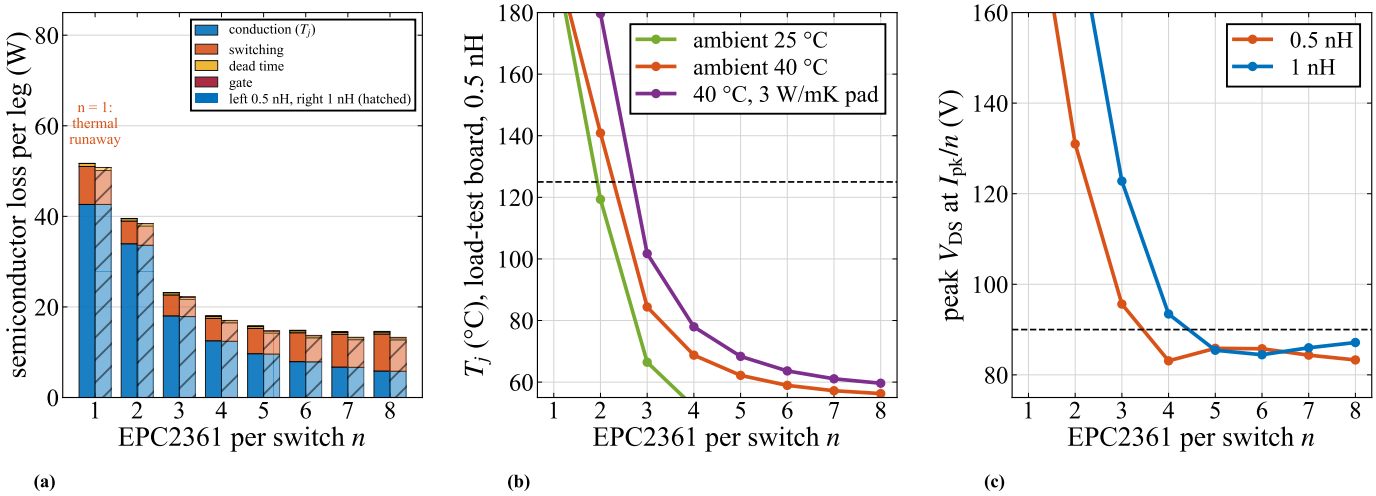


Fig. 3. n EPC2361 per switch at 228 $\mathrm{A_{rms}}$ and 50 kHz. (a) Semiconductor loss per leg, 0.5 nH (left) and 1 nH (right, hatched); (b) junction temperature on the load-test heatsink; (c) peak V_{DS} at the peak current per device.

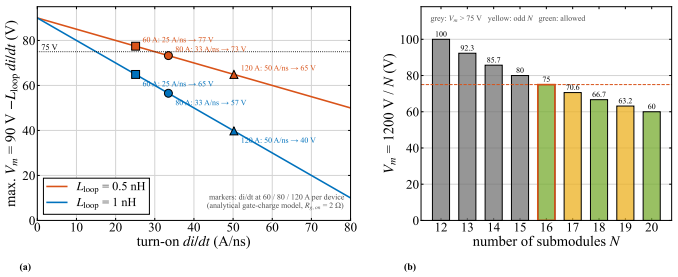


Fig. 4. (a) Admissible submodule voltage $90\,\mathrm{V} - L_{\mathrm{loop}}\,di/dt$ for 0.5 and 1 nH; markers: turn-on di/dt at 60, 80 and 120 A per device. (b) $V_m = 1200\,\mathrm{V}/N$; green: admissible and even.

50 kHz. A ripple of 3 % (2.25 V) therefore requires 127 µF effective capacitance per leg board; the voltage ripple, not the current, sizes the dc link. At 75 V, 100 V class-II MLCCs retain only 15–29 % of their nominal capacitance. The 22 μF 2220 part with the highest retention (29 %, 6.35 μF) is selected, and twenty of them are placed on each leg board (Fig. 7); the ripple current alone would need only 7 parts.

Ten 1 μF 0805 capacitors per cell, on both board sides at the high-side drains, close the commutation loop.

VI. Leg Board and Verification

The four-layer leg board (2 oz, 0.1 mm dielectric between L1 and L2) carries the four cells as two identical pairs, each with its driver in the centre between two cells rotated by 180° (Fig. 6). Each commutation loop is vertical: the forward path on L1 returns on the dc− plane on L2 directly below. One dc terminal pair feeds a laminated row across the board that supplies both the capacitor bank and all cells, and the four switch nodes are joined in copper and leave through a single bolted ac terminal; 15 vias per cell tie each switch-node island to the ac copper on the lower layers. Symmetry required two measures. The interleaved drain and source stripes of the

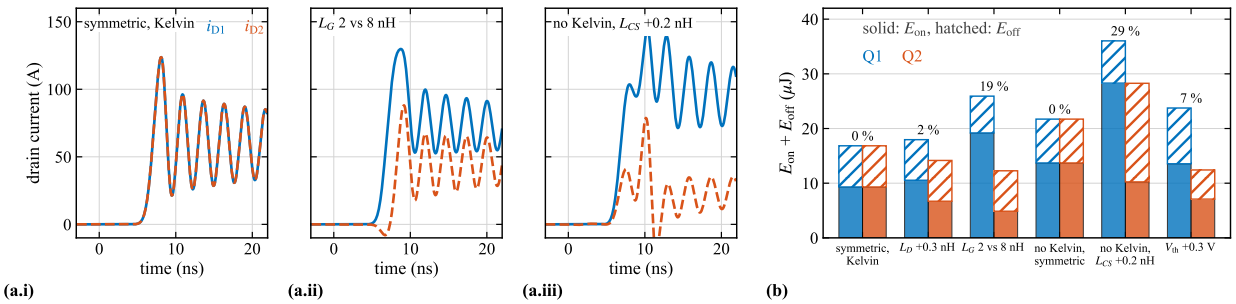


Fig. 5. Current sharing of two paralleled EPC2361 (LTspice, 75 V, 120 A). (a) Drain currents at turn-on: (i) symmetric with Kelvin connections, (ii) gate loops of 2 and 8 nH, (iii) no Kelvin connection and 0.2 nH common-source mismatch. (b) Turn-on (solid) and turn-off (hatched) energy of both devices.

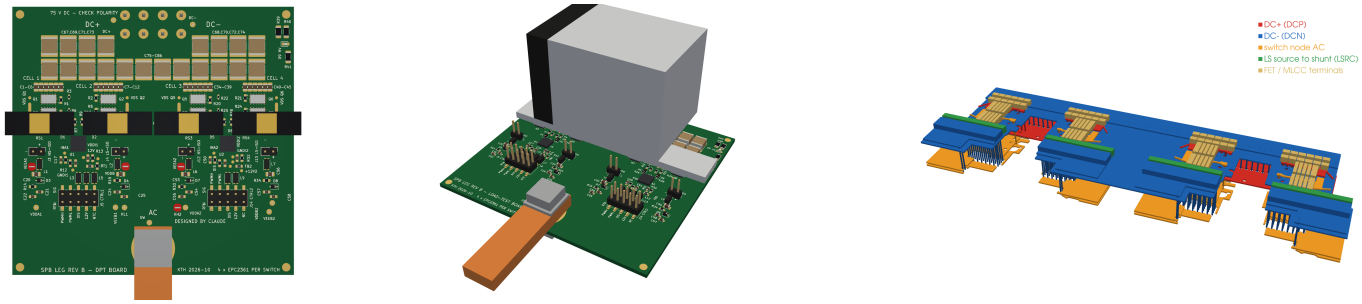


Fig. 6. Leg board, four EPC2361 per switch (89.5 × 85.2 mm, four layers). (a) Double-pulse-test board with a coaxial shunt per cell; (b) load-test board with fan heatsink; (c) Q3D model of the cells of the test board, generated from the routed copper (red dc+, blue dc−, orange switch node; z stretched).

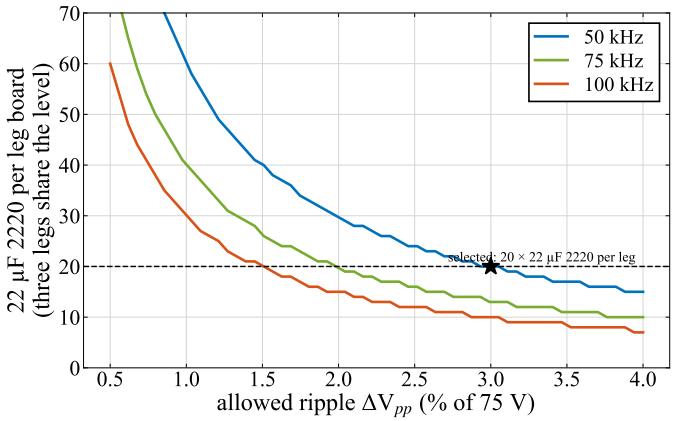


Fig. 7. Number of 22 μF 2220 MLCCs per leg board against the peak-to-peak ripple of the 75 V level, including the dc-bias derating; star: selected design.

TABLE II
Q3D Results of the Two Boards

	DPT	Load test
L_{loop}, cell 1/2/3/4 [nH]	0.646/0.644/ 0.644/0.648	0.408/0.408/ 0.406/0.412
Copper / bypass ESL / shunt [nH]	0.46/0.06/0.12	0.35/0.06/–
$\hat{v}_{DS}$ at 30 A/ns [V]	94	87
Cell-to-cell M [pH] / k, max.	20/0.033	–
L_G HS/LS, alone; pair [nH]	12.5/7.6; 20.6/12.3	–
Static imbalance, HS/LS [%]	**TBD/TBD**	**TBD/TBD**
Board copper loss [W]	**TBD**	**TBD**

EPC2361 leave no room for a via row, so the low-side source stripes use vias in the pad, and any via removed for clearance is removed together with its mirror image. The driver inputs (common-mode chokes on supply and disable, none on PWM) keep 4.0 mm on the outer and 3.0 mm on the inner layers from all power copper, the IPC-2221B spacing for 1200 V, so the layout is ready for a reinforced isolated driver; the 2EDF7275K used here provides functional isolation for the 75 V tests.

A shunt in every low-side source is needed to measure the cell currents but lengthens the loop and blocks the heatsink. The board is therefore built in two versions from the same design. The double-pulse-test (DPT) board carries a 1 mΩ coaxial shunt (PMK UFCS) per cell on a shortened land and needs no heatsink; MMCX jacks and spring-probe pads give v_{DS} and v_{GS} of every device. The load-test board connects the low-side sources through blind vias to the dc− plane and carries the heatsink.

The routed boards were verified in Ansys Q3D with the loop evaluated with all four cells switching. To keep the solution time short, each quantity is extracted from the smallest model that contains it: the cell band at 100 MHz for the loop and the full board at dc for the resistances. Table II and Fig. 8 summarize the results. The load-test board reaches 0.41 nH per cell, of which 0.35 nH is copper and 0.06 nH capacitor ESL,

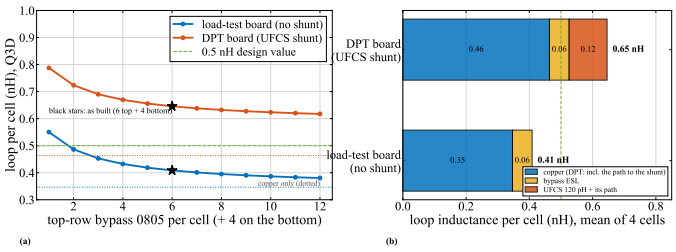


Fig. 8. Loop inductance per cell (Q3D). (a) Against the number of top-side bypass capacitors (stars: as built); (b) contributions of copper, capacitor ESL and shunt.

TABLE III
COMPARISON WITH PARALLELED-GAN HALF BRIDGES

Work	Devices per switch	Operating point	L_{loop}
[12]	4×650 V	400 V, 240 A (DPT)	0.7 nH
[13]	4×100 V	48 V, 328 A (DPT)	1.14 nH
[15]	8×100 V	48 V, 125 A_{rms}	–
[16]	8×200 V	120 V, 155 A	0.34 nH*
This work	4×100 V	75 V, 228 A_{rms}	0.41 nH/cell

*analytical estimate

and the four cells agree within 1.6 %. It thus meets the 0.5 nH target and gives about 87 V peak at 30 A/ns. On the DPT board the shunt and its connection add 0.24 nH, which costs overshoot margin (Section VII). Beyond four bypass capacitors per cell the loop hardly decreases (Fig. 8a); the remaining capacitors provide capacitance rather than lower inductance. The cells hardly couple (at most 20 pH, $k \leq 0.033$, between the inner cells), so each cell commutates through its own loop.

A second model adds the gate network of one driver block (DPT board); gate and Kelvin return run on adjacent layers from each gate pad to the output capacitors of the driver. A device driven alone sees 12.5 nH (high side) and 7.6 nH (low side). The star-to-driver part (8.0 and 4.7 nH) is shared, so a device of a pair sees 20.6 and 12.3 nH. All four paralleled devices of a switch agree within 0.3 %, and the commutation loops couple at most 9 pH into a gate loop. The gate loop is thus about ten times the lumped 1.7 nH, because the driver sits between two cells behind the isolation barrier.

Three load-test boards form a level on a four-layer laminated PCB busbar (Fig. 10). The busbar only exchanges the switching-frequency ripple between the capacitor banks of the three legs. The inductance between neighbouring legs is 1.46 nH (1.98 nH between the outer legs at 1 MHz), small against the reactance of a bank at 50 kHz, so the banks share the ripple. Sixteen such levels in series form the 1200 V inverter.

Table III places the design among paralleled-GaN half bridges. With four parallel cells, each with its own loop of 0.41 nH, the leg reaches a lower loop inductance than the module-type layouts [12], [13], while the per-pair gate drive addresses the sharing mechanisms reported for them. The remaining limits are the overshoot margin, set by the turn-on speed, and the thermal path through the insulating gap pad, which an integrated design would replace by a ceramic-insulated path [13].

VII. VALIDATION WITH THE EXTRACTED PARASITICS

The device-level design of Sections III and IV relies on a lumped loop. A routed board adds what this model hides: the four cells reach the capacitor bank and the ac terminal through different copper, couple to each other, and see their bypass capacitors at different distances. To check the design against these effects, the extracted copper was inserted into the double-pulse circuit of Section IV: one inductor per terminal with every mutual inductance as a coupling (29 inductors and 400 couplings for the load-test board), the dc resistance matrix through behavioral sources, eight EPC2361 models, the bypass capacitors with dc-bias capacitance and ESL, the shunts, the bank and two drivers with star gate networks. The load current is switched at the peak phase current.

Fig. 9 shows the result. The four cells carry 77–81 A at turn-off and their turn-on energies differ by 2 % (0 % on the DPT board), so the mirrored layout and the per-pair gate drive leave none of the sharing mechanisms of Table I active. The low-side turn-off is uncritical (81 V). The critical event is the low-side turn-on, at which the high-side device is charged at 46 A/ns. Its drain–source voltage peaks at 91 V on the load-test board, in agreement with the lumped model at the extracted loop (91 V), and at 93 V on the DPT board. The energies agree with the lumped model within 10 % for turn-on (27.3 versus 29.8 µJ) and within 1 µJ for turn-off. The peak gate bump of the high-side devices, 1.2 V with a 0 V off-state, exceeds the 0.8 V minimum threshold and confirms the need for the negative off-state voltage of Section IV.

The simulation therefore validates the lumped design of the power loop but shows that the load-test board operates exactly at the 90 V limit with $R_{\text{g,on}} = 2\,\Omega$. With 3 Ω the turn-on slows to 34 A/ns and the peak drops to 86 V. The turn-on energy rises by 20 % (32.8 µJ per device), which adds less than 1 W per leg.

The gate loop is the exception. With the extracted gate network of the DPT board (plus the resistor and diode bodies) and $R_{\text{g,on}} = 2.5\,\Omega$, the turn-on reaches 49 A/ns and the high-side peak 105 V (90 V with the lumped gate loop). The DPT board is therefore driven with 3.5 Ω (1.5 Ω shared per pair) and -1.9 V: the peak is 93 V with the shunts in the loop, the turn-on energies of the cells agree within 0 %, and the held-off gate peaks at -0.5 V, 1.3 V below the minimum threshold, while the off-state ringing (about -3.4 V) stays inside the -4 V limit.

VIII. CONCLUSION

Because the phase current of an SPB does not depend on the number of submodules, the device and the submodule voltage can be chosen for the lowest loss and the available overshoot margin: for 1200 V and 300 kW, 16 submodules of 75 V with four 100 V GaN transistors per switch, driven in pairs by one isolated dual-channel driver each. Q3D extraction of the routed board gives 0.41 nH per cell, equal loops in all four cells and negligible coupling between them ($k \leq 0.033$). A double-pulse

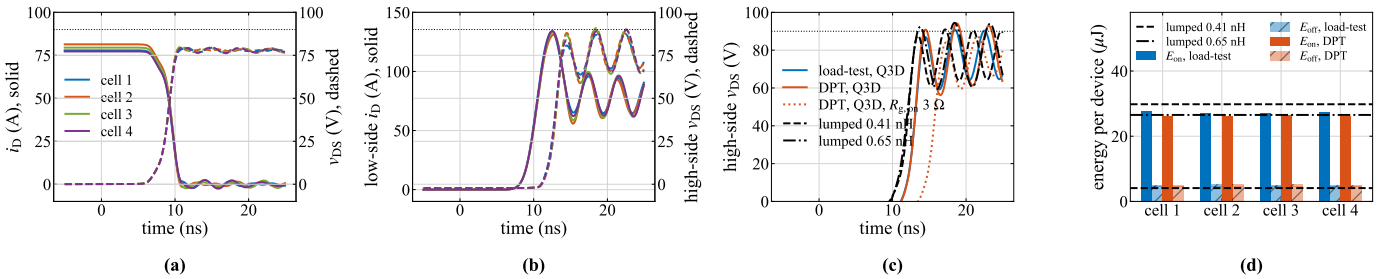


Fig. 9. Double-pulse simulation with the Q3D-extracted copper, 75 V, 316 A load current (≈ 80 A per device), $R_{g,on} = 2\,\Omega$. (a) Low-side turn-off and (b) low-side turn-on on the load-test board, all four cells: drain current (solid) and v_{DS} (dashed). (c) High-side v_{DS} during the low-side turn-on, cell with the highest peak, compared with the lumped model at the extracted loop inductance. (d) Switching energy per device in each cell; lines: lumped model.

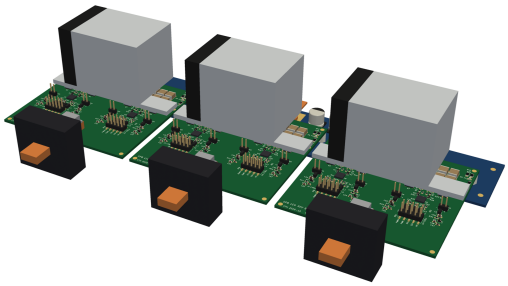


Fig. 10. One 75 V level: three load-test boards on the laminated PCB busbar, ac bars through Hall-effect current transducers.

simulation with the full extracted parasitic network confirms the lumped design of the power loop: the cells share current within 3 % and the switching energy within 2 %, and a slightly slower turn-on ($R_{g,on} = 3\,\Omega$) keeps the peak voltage at 86 V. The gate loop of the centre driver, equal for all four paralleled devices but about ten times the assumed value, is the main deviation: the test version with shunts (0.65 nH) needs 3.5 Ω and −1.9 V to stay at 93 V. A driver stage next to each pair of devices is planned for the next revision. Double-pulse and continuous tests of a three-phase level are the next step.

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